

THE $n = 5$ CASE OF LENTFER'S (1, 2)-BOSONIC-FERMIONIC COINVARIANT BASIS CONJECTURE

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ABSTRACT. Lentfer conjectured a monomial basis $B_n^{(1,2)}$ for the (1, 2)-bosonic-fermionic coinvariant ring and reported verification for $n \leq 4$. We prove the first finite case beyond that reported range: the 1920 residue classes in $B_5^{(1,2)}$ form a basis. First we give a uniform structural reduction showing that the ideal generated by all positive-degree invariants is generated by $4n$ polarized power sums. We then certify the $n = 5$ claim by exact standard-basis and rational-rank computations. A standalone, fail-closed verifier reconstructs the candidate set and the invariant ideal from serialized data; it also reproduces the known $n = 3, 4$ cases. The result is a certified finite theorem and does not prove the conjecture for arbitrary n .

1. INTRODUCTION

Let S_n act diagonally on one set of commuting variables and two sets of anticommuting variables. Lentfer constructed a finite monomial set $B_n^{(1,2)}$, conjectured that it is a basis of the corresponding coinvariant ring, proved that its cardinality is $2^{n-1}n!$, and reported that the conjecture had been verified for $n \leq 4$ [1, Definition 3.1, Conjecture 3.2, Remark 3.3, Theorem 3.7]. Thus $n = 5$, where the candidate has $2^4 5! = 1920$ monomials, is the first case not covered by the verification reported there.

Our result settles this finite case.

Theorem 1.1. *Over every field of characteristic zero, the residue classes of $B_5^{(1,2)}$ form a basis of $R_5^{(1,2)}$. In particular,*

$$\dim R_5^{(1,2)} = 1920.$$

The proof has a conceptual and a certified computational part. The conceptual part replaces the apparently infinite invariant ideal by an exact family of $4n$ generators. The computational part works over $\mathbb{Q}$, constructs a standard basis in a supercommutative algebra, and checks that the normal forms of the 1920 candidates have rational rank 1920. All data and commands needed to reconstruct this finite statement accompany the paper.

The declared degree-product term order is only an implementation choice. In that order the candidates are not literally all standard monomials. What

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is proved is the source conjecture's order-independent basis assertion: their normal forms are a basis of the quotient.

2. DEFINITIONS AND REDUCTIONS

Fix a characteristic-zero field K and define

$$A_n = K[x_1, \dots, x_n] \otimes_K \Lambda_K(\theta_1, \dots, \theta_n, \xi_1, \dots, \xi_n).$$

The variables x_i are even. All $2n$ odd variables anticommute with one another, including across the two species. We use the canonical order

$$\theta_T \xi_S = \theta_{t_1} \cdots \theta_{t_p} \xi_{s_1} \cdots \xi_{s_q} \quad (t_1 < \cdots < t_p, \quad s_1 < \cdots < s_q).$$

The action is $\sigma(x_i) = x_{\sigma(i)}$, $\sigma(\theta_i) = \theta_{\sigma(i)}$, and $\sigma(\xi_i) = \xi_{\sigma(i)}$. If $(A_n^{S_n})_+$ denotes the positive-degree invariants, then

$$I_n = \langle (A_n^{S_n})_+ \rangle, \quad R_n^{(1,2)} = A_n / I_n.$$

We recall the candidate in a form convenient for verification. A modified Motzkin path starts with an up-step and thereafter stays at height at least one. Its four steps are up, θ -horizontal, ξ -horizontal, and down, with respective weights $1, \theta_i, \xi_i, \theta_i \xi_i$. For a path of weight $\theta_T \xi_S$, define

$$\alpha_1(T, S) = 0, \quad \alpha_i(T, S) = \alpha_{i-1}(T, S) - 1 + \mathbf{1}_{i \notin T} + \mathbf{1}_{i \notin S} \quad (2 \leq i \leq n).$$

Then $B_n^{(1,2)}$ consists of all

$$x_1^{a_1} \cdots x_n^{a_n} \theta_T \xi_S, \quad 0 \leq a_i \leq \alpha_i(T, S),$$

over the modified Motzkin path weights $\theta_T \xi_S$. A step-ordered odd weight can differ from this species-ordered canonical form by a sign; rescaling an individual vector by -1 does not affect the basis assertion. With this canonicalization, the description is equivalent to Lentfer's definition [1, Definition 3.1].

For $U \in \{1, \theta, \xi, \theta\xi\}$, put

$$p_{r,U} = \sum_{i=1}^n x_i^r U_i, \quad U_i \in \{1, \theta_i, \xi_i, \theta_i \xi_i\}.$$

Proposition 2.1 (finite invariant generators). *For every $n \geq 1$,*

$$I_n = \langle p_{r,1} \ (1 \leq r \leq n), \quad p_{r,U} \ (0 \leq r < n, \ U \in \{\theta, \xi, \theta\xi\}) \rangle.$$

Proof. Let $P = K[x] \otimes \Lambda(\theta, \xi)$, so that $A_n = P^{\otimes n}$ with the super place-permutation action. For a homogeneous $a \in P$, write $p(a) = \sum_i a^{(i)}$.

For homogeneous positive-degree monomials

$$a_1, \dots, a_k \in P,$$

let $M(a_1, \dots, a_k)$ be the signed sum over injections of these factors into distinct tensor slots, with the Koszul sign obtained on returning the product to increasing slot order. Each Reynolds average of a monomial is a scalar multiple of such a sum. The sum may be zero when an odd stabilizer acts

by -1 , but Reynolds images of the ambient monomial basis still span the invariant subspace. Thus the (possibly zero) M 's span the invariants.

Expanding $p(a_1) \cdots p(a_k)$ and grouping slot maps by their kernel partitions gives

$$p(a_1) \cdots p(a_k) = M(a_1, \dots, a_k) + \sum_{\substack{\pi \text{ a partition of } [k] \\ |\pi| < k}} \varepsilon_\pi M((a_B)_{B \in \pi}).$$

Within a block B , the product a_B is taken in increasing label order; ε_π is the corresponding Koszul sign. Odd collisions may make a summand zero, but the discrete-partition term always has coefficient one. Induction on k therefore shows that the invariant algebra is generated by the one-slot sums $p(a)$. The one-slot monomials are exactly $x^r, x^r\theta, x^r\xi, x^r\theta\xi$, with $p(1) = n$ excluded as a scalar.

It remains to truncate r . Let $e_j = e_j(x_1, \dots, x_n)$. Newton's identities give

$$\langle p_{1,1}, \dots, p_{n,1} \rangle = \langle e_1, \dots, e_n \rangle.$$

Since each x_i is a root of $\prod_j (t - x_j)$, multiplying its characteristic identity by $x_i^{r-n} U_i$ and summing gives, for $r \geq n$,

$$p_{r,U} = e_1 p_{r-1,U} - e_2 p_{r-2,U} + \cdots + (-1)^{n+1} e_n p_{r-n,U}.$$

Thus every higher one-slot sum belongs to the displayed ideal. Conversely, each displayed generator is a positive-degree invariant. The two ideals are equal. $\square$

3. MAIN RESULT

The computation uses the $(\deg_x, \deg_\theta, \deg_\xi)$ -grading. Candidate generation is literal: masks (T, S) are enumerated, the path height is checked after every step, the generalized α -sequence is reconstructed, and every bounded exponent tuple is emitted. The resulting counts are 24, 192, 1920 at $n = 3, 4, 5$.

For each n , Proposition 2.1 supplies the exact input ideal. Over $\mathbb{Q}$, a two-sided standard basis G_n is computed in the supercommutative quotient. Let K_n be its standard-monomial basis. The exact normal forms of all candidate monomials are expanded in K_n , giving a coordinate matrix M_n . The certified outputs are as follows.

n	generators	candidates	$\dim A_n/I_n$	$\text{rank } M_n$	$ G_n $
3	12	24	24	24	19
4	16	192	192	192	59
5	20	1920	1920	1920	271

Proof of Theorem 1.1. All 20 polynomials used at $n = 5$ generate I_5 by Proposition 2.1. Exact standard-basis reduction gives $\dim_{\mathbb{Q}} A_5/I_5 = 1920$. The independently reconstructed candidate list has 1920 distinct elements, and the exact coordinate matrix of their normal forms has rank 1920. Their

residue classes are therefore linearly independent and, by dimension, form a basis over $\mathbb{Q}$. Rank and dimension are preserved by scalar extension to any characteristic-zero field. $\square$

4. CERTIFICATE OR COMPUTATIONAL VERIFICATION

The serialized certificate records the field, dimension, term order, all $4n$ sparse invariant generators, every candidate monomial, and its multidegree block. The verifier imports no discovery module. It instead:

- (1) reconstructs the modified Motzkin paths and checks the serialized candidate list term by term;
- (2) reconstructs the $4n$ family of Proposition 2.1 and checks the sparse input term by term;
- (3) recomputes G_n , verifies that every input generator reduces to zero, and obtains K_n ;
- (4) checks that $K_n M_n$ exactly reconstructs every candidate normal form;
- (5) computes $\dim A_n/I_n$ and $\text{rank } M_n$ over $\mathbb{Q}$, rejecting unless both equal $2^{n-1}n!$.

Independent block audit. A second implementation avoids the polarized-power-sum standard basis. It first passes to the finite Artin–exterior algebra

$$C_n = (\mathbb{Q}[x_1, \dots, x_n]/(e_1, \dots, e_n)) \otimes \Lambda(\theta_1, \dots, \theta_n, \xi_1, \dots, \xi_n).$$

Equivariance and Reynolds averaging give

$$R_n^{(1,2)} \cong C_n / \langle (C_n^{S_n})_+ \rangle.$$

The audit reconstructs every invariant block by signed Reynolds sums. It then closes the ideal recursively under the $3n$ algebra generators. Exact rational row reduction in python-flint 0.9.0 checks all 396 nonempty tridegrees at $n = 5$: the ambient dimension is 122880, the total invariant-space dimension is $1024 = 4^5$, the quotient dimension is 1920, and in every block the candidate vectors are both spanning and independent modulo the ideal. This route imports neither the standard-basis discovery code nor its serialized candidate file.

Malformed input, a duplicated candidate, a changed generator coefficient, a wrong multidegree count, or a missing schema field is rejected. Five such acceptance/rejection tests accompany the verifier. The decisive serialized $n = 5$ input has SHA-256 digest

09700e99a7add186ea0f04877b33f041b
cd9e01c8633cf83aa15131f596bf25f.

The exact engine used for the recorded run was Singular 4.4.1 (build 44105) over $\mathbb{Q}$; the executable and the supercommutative relations are declared by the verifier. The one-command reproduction is

`sh experiments/run_exact.sh`

which also repeats the $n = 3, 4$ baselines and prints all code and input hashes.

5. EQUALITY CASES AND LIMITATIONS

There are no optimization equality cases. The endpoint is finite and exact: it proves Conjecture 3.2 only for $n = 5$ (while reproducing $n = 3, 4$). It does not establish a recurrence for general n , and it does not assert that $B_5^{(1,2)}$ is the standard-monomial set for the declared term order. In that order, 1488 of the 1920 candidates are already normal; the remaining 432 reduce to nontrivial exact linear combinations, while all 1920 normal forms remain independent.

DATA, CODE, AND COMPUTATIONAL DISCLOSURE

The certificate, verifier, tests, source, hashes, and clean-run command are provided with this manuscript. The verifier uses no random seed and no floating-point or prime-field calculation. Artificial-intelligence tools assisted with literature search, proof exploration, implementation, and drafting. The mathematical endpoint is supported by the displayed proof of the invariant-generator reduction and by exact, independently rerunnable certificate checks. No Lean, Coq, Isabelle, or other proof assistant was used.

REFERENCES

1. John Lentfer, *A conjectural basis for the $(1, 2)$ -bosonic-fermionic coinvariant ring*, Algebraic Combinatorics **8** (2025), no. 3, 711–743 (en).

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